

Industry-Oriented Detection Method of PCBA Defects Using Semantic Segmentation Models

Yang Li , Xiao Wang , Zhifan He , Ze Wang , Ke Cheng , Sanchuan Ding , Yijing Fan ,
Xiaotao Li , Yawen Niu , Shanpeng Xiao , Zhenqi Hao , Bin Gao ,
Senior Member, IEEE, and Huaqiang Wu , Senior Member, IEEE

Abstract—Automated optical inspection (AOI) is a significant process in printed circuit board assembly (PCBA) production lines which aims to detect tiny defects in PCBAs. Existing AOI equipment has several deficiencies including low throughput, large computation cost, high latency, and poor flexibility, which limits the efficiency of online PCBA inspection. In this paper, a novel PCBA defect detection method based on a lightweight deep convolution neural network is proposed. In this method, the semantic segmentation model is combined with a rule-based defect recognition algorithm to build up a defect detection framework. To improve the performance of the model, extensive real PCBA images are collected from production lines as datasets. Some optimization methods have been applied in the model according to production demand and enable integration in lightweight computing devices. Experiment results show that the production line using our method realizes a throughput more than three times higher than traditional methods. Our method can be integrated into a lightweight inference system and promote the flexibility of AOI. The proposed method builds up a general paradigm and excellent example for model design and optimization oriented towards industrial requirements.

Index Terms—Automated optical inspection (AOI), deep learning, defect detection, printed circuit board assembly (PCBA), semantic segmentation.

I. INTRODUCTION

WITH the rapid development of the electronics industry, the demand for printed circuit boards (PCBs) is increas-

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Y. Li, X. Wang, Y. Fan, X. Li, Y. Niu, and S. Xiao are with the Department of Internet of Things Technology and Application, China Mobile Research Institute, Beijing 100053, China (e-mail: liyangwx@chinamobile.com; wangxiaow1@chinamobile.com; fanyijing@chinamobile.com; lixiaotao@chinamobile.com; niuyawen@chinamobile.com; xiaoshanpeng@chinamobile.com).

Z. He was with the Beijing Advanced Innovation Center for Integrated Circuits, the School of Integrated Circuit, Tsinghua University, Beijing 100084, China. He is now with the SigmaStar Technology Limited, Xiamen 361199, China (e-mail: hzf20@mails.tsinghua.edu.cn).

Z. Wang, K. Cheng, S. Ding, Z. Hao, B. Gao, and H. Wu are with the Beijing Advanced Innovation Center for Integrated Circuits, the School of Integrated Circuit, Tsinghua University, Beijing 100084, China (e-mail: wangze22@mails.tsinghua.edu.cn; ck20@mails.tsinghua.edu.cn; zibaizhenyun@tsinghua.edu.cn; zqhao@tsinghua.edu.cn; gaobl1@tsinghua.edu.cn; wu hq@tsinghua.edu.cn).

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ing. Therefore higher performance is required for the automatic and intelligent manufacturing of PCBs. Surface mount technology (SMT) is mainstream technology applied in PCB automated manufacturing. SMT combines a sequence of working steps to automatically output final products which are called printed circuit board assembly (PCBA). Nowadays, automated SMT production lines have become the standard configuration in PCB manufacturing, which can greatly reduce labor costs and improve production efficiency. An SMT production line consists of several steps including board loading, dispensing, soldering, defect inspection, and board unloading, etc. Among all these steps, defect inspection is critical and it determines the production yield of PCBAs.

Currently, automatic optical inspection (AOI) equipment is widely applied in automated SMT production lines. AOI combines optical technology and pattern recognition to detect tiny defects in semi-finished PCBAs. As shown in Fig. 1, usually AOI includes two stages which are image acquisition and defect detections respectively. During the image acquisition stage, original images of PCBA are obtained through an industrial camera and light source. While in the defect inspection stage, original images are analyzed and detected by the calculation and inference module integrated into AOI, and inspection results are then output to the screen. In the existing production lines, modules in charge of these two stages are integrated into one piece of equipment. Thus, the equipment is relatively large in size and power consumption, especially when the storage device is put into consideration. Moreover, the inspection module has to be readjusted when the size of the PCBA changes, making the production line poor in flexibility.

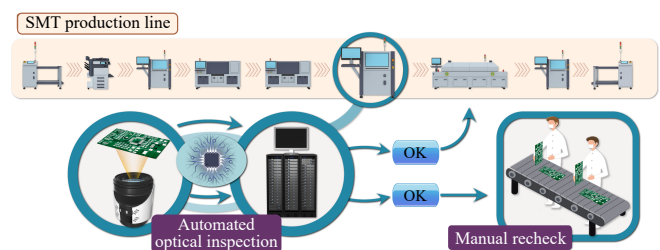


Fig. 1. Schematic of PCBA defect detection process on SMT production line.

Nowadays under the trend of flexible manufacturing, the ability to adapt to several types of products is urgently needed. Thus, the importance of the edge intelligent system (EIS) is

increasing rapidly in modern PCB production lines. An EIS is a lightweight computing device embedded with an intelligence chip for model inference and some other functional modules. With the help of an EIS, the inference model can be decoupled from the AOI equipment and conveniently edited according to different working conditions. However, a problem with applying an EIS in the detection process is that its small scale limits the computing power. With limited computing power, existing methods are not able to ensure the accuracy and efficiency of PCBA defect detection. Therefore, a challenging task we are faced with is to explore a novel PCBA defect detection method which can be integrated into EIS and simultaneously promote detection performance in actual production lines.

Presently, several methods of PCBA defect detection have been proposed. These methods can be roughly divided into two categories, namely the template-based method and template-free method respectively. Template-based method is a traditional method, which mainly compares the parameters of the PCB image pixel by pixel with the reference template to find the abnormal positions [1]–[3]. This method proposes strict requirements for template quality. Template-free methods, which have been developing rapidly in recent years, are mainly based on object detection (OD) frameworks. In these methods, a convolution neural network (CNN) is used to directly infer the locations and types of different defects from the original images, instead of using the reference template [4]–[17]. Recent research has proved that deep learning-based methods can effectively solve the PCBA defect detection problem and improve detection accuracy compared with traditional methods. However, there are still several problems when applying these methods to lightweight computing devices. First, the models used in previous studies are large in scale which are difficult to deploy with limited computing and storage resource. Even if the reduced models can be deployed in EIS, they can hardly meet the real-time requirements of industrial site. Secondly, in previous studies, open-source PCB defect datasets were used to train the models. However, the existing data set only contains simple PCB images and some of the defects are generated manually. Thus, the trained models show little generalization capability for complex PCBA products. Therefore, it is necessary to propose a defect detection method for complex PCBAs which can be compatible with EIS and operate efficiently in modern intelligent production lines.

This paper proposes a novel defect detection method for complex PCBAs which is geared towards actual production lines. The method uses the CNN model to extract segmentation maps of raw PCBA images and detect tiny defects using visual geometric features. The main contributions of this article are as follows:

- 1) A framework based on semantic segmentation is proposed for PCBA defect detection which can successfully detect defects in PCB with different designs but the same color in modern SMT production lines.
- 2) We build up a general method for model design and optimization oriented to constraints from PCBA production lines. Under this method, the detection model can be integrated into

lightweight computing devices and applied in different working conditions.

- 3) A PCBA data set for defect detection is built based on the images obtained in SMT production lines. The data set covers the main defect types in the actual production of PCBA and has been annotated manually.

The rest of this article is organized as follows. Section II reviews previous related research. In Section III, we introduce the defect detection details and present an overview of the AOI inspection system. Section IV describes the details of the semantic segmentation model and other optimization algorithms. In Section V, we train the proposed model and experimentally evaluate the proposed method through a test system production line. Finally, we summarize our research and propose the direction of our future work.

II. RELATED WORK

A. PCB Defect Detection

According to template usage in the inspection process, the existing studies on PCB defect detection are mainly divided into two categories, which are template-involved (TI) and template-free (TF), respectively. The TI method mainly compares the PCB image to be tested with the reference template to determine the positions of existing defects. With the assistance of a template, even some tiny and infrequent defects can be found. Early TI methods involve pixel-wise subtraction to get a difference map [1]–[3]. However, this kind of method requires that the test image aligns well with the template.

TF methods are a series of techniques in PCB defect detection which help directly recognize defects without the assistance of a template. In recent years, deep learning has been mainstreamed in the field of object detection, making TF methods popular in PCB defect detection. TF methods originate from two-stage object detection algorithms such as R-CNN [18], Fast-RCNN [19], Faster-RCNN [20] and Mask-RCNN [21]. In such algorithms, a CNN is utilized as the backbone to extract features of the original PCB image, while regions of interest are then chosen by the region proposal network (RPN). For example, Ding *et al.* proposed a defect detection model that combines Faster-RCNN with multi-scale feature fusion and used open source PCB defect data set to train the model [4]. Hu and Wang propose another Faster-RCNN based approach which combines ResNet50 with FPN as a feature extractor and uses GARP to improve the accuracy of bounding box regression [5].

B. Image Segmentation Algorithms

Semantic segmentation (SS) is a classic computer vision problem which involves taking raw data as input and converting them into a mask with a highlighted area of interest. In segmentation, each pixel in the image is assigned a category ID according to the object it belongs to. In this way, semantic segmentation brings a pixel-level understanding of images and expands its application fields. With the development of deep learning, neural networks have been a main approach to realizing SS. Deep learning-based segmentation originates from a fully convolution network which performs pixel-wise classifi-

cation with an end-to-end method [22]. After that, a series of improvements on model structure have been proposed to optimize the performance of semantic segmentation [23]–[26]. These improvements include enlarging the receptive field, refining the contextual information, introducing boundary information, and designing a multi-scale attention module. For example, U-Net is proposed to accomplish precise segmentation in medical images which have refined the biological structure [23]. Deeplab utilizes Atrous convolution and fully connected conditional random fields (CRFs) to solve problems in deep learning-based segmentation, such as with feature resolution decreases and multi-scale objects [25]. However, these improved models bring improvement in segmentation performance at the expense of extra complexity and computational resources. Semantic segmentation is highly consistent with the task of PCBA detection because the minor structure of PCBA requires a classification method on a micro scale. Nevertheless, the balance of segmentation performance and computation demand is still a challenging task before an SS model is well used in PCBA defect detection.

III. PCBA DEFECT DETECTION SYSTEM

A. PCBA Defects

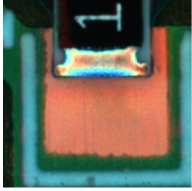
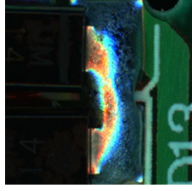
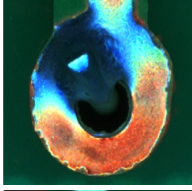
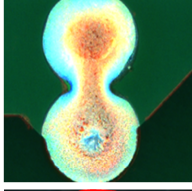
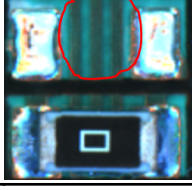
In SMT production lines, wave soldering is a significant process link in which over 10 types of possible soldering defects may come into existence. Normally, we can detect these defects during the post-soldering inspection. However, there is no comprehensive AI-based solution for post-soldering inspection, and not a dataset which can be used for model training.

According to the statistics from actual production lines, we find that there are five representative types of defects occurring frequently in most SMT components (small outline package, small outline diode, etc.) which are the bridge, empty, pinhole, pin missing, and chip missing defects. In this paper, our method is mainly designed for detection of these five defects during post-soldering inspection. As shown in Table I, each category of defect shows corresponding visual characteristics which may differ hugely in geometry. Moreover, our method can also be applied to other defects, such as chipping, scratch, and contamination with enough training data.

B. AOI Framework

The AOI framework proposed in this article mainly consists of two parts, namely the PCBA image acquisition system (PCBA-IAS) and inference system as mentioned before. In a detection cycle as shown in Fig. 2, PCBA images are initially obtained by an industrial camera with the coaxial light source. These images are then transmitted to the inference system and pre-processed to eliminate redundant information. Afterwards, the pre-processed images are inferred by the proposed model deployed in a computing device and transferred into segmentation maps. In segmentation maps, soldering areas are divided from the background. By comparing the segmentation with a template, the defects are recognized and displayed on the screen, after which the command is sent to the subsequent actuator to pick out the defected PCBA.

TABLE I
FIVE MAIN DEFECT TYPES OF PCBA IN SMT PRODUCTION LINES

Defect type	Image example	Description
Opens		An open connection between the lead and pad. This usually happens when the lead and the pad don't bond, or when the solder is only on the PCB but not on the component lead.
Solder bridge		Solder improperly connects one lead to another. Solder bridges are very hard to see and can cause serious damage to circuits assembly if undetected.
Pin hole		Holes in solder point caused by bubbles or other effects. This may lead to reliability failure during the component I/O test.
Pin missing		Pins on solder joints are not sticking out.
Components missing		Electronic components are missing in target areas. This leads to a loss of PCBA functionality.

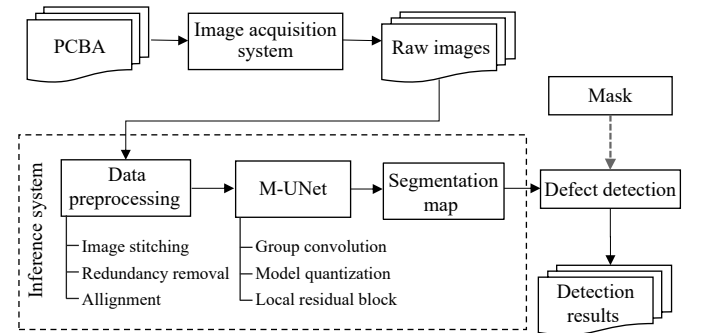


Fig. 2. Sketch of proposed PCBA defect detection method.

C. Image Acquisition System

The image acquisition system (IAS) consists of a CMOS camera and a coaxial light source. The coaxial light source, consisting of a high-density LED array, provides different illumination angles and different color combinations which can better highlight the three-dimensional information of PCBA to be inspected. As shown in Fig. 3, the camera moves on the guide rails and starts to take pictures when PCBA reaches right above it. To cover the entire area of PCBA, the camera moves in a zigzag trajectory and takes a total of

20 pictures with the size of 2448×2048 for one PCBA with a 17 cm by 12 cm size. All these 20 pictures are stitched together to form a complete input image of the inference system.

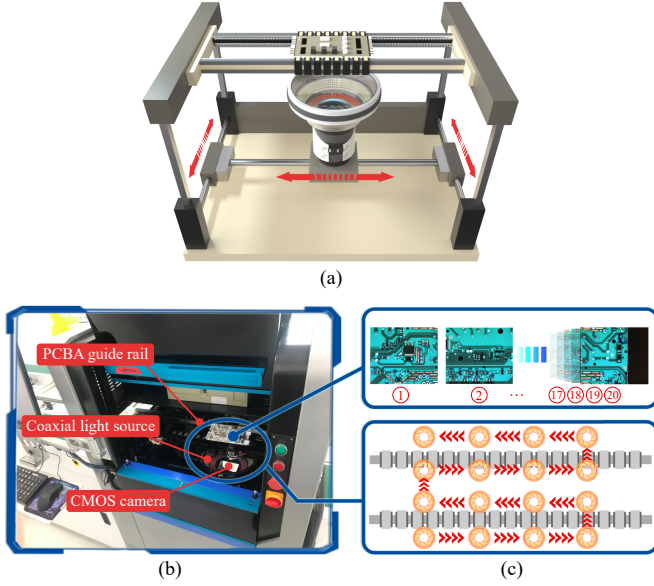


Fig. 3. Image acquisition system of the PCBA defect detection. (a) Schematic of AOI equipment; (b) An example of AOI equipment in the actual production line; (c) Top: Samples of raw images obtained by the camera; Bottom: Motion trajectory of CCD camera when taking pictures of PCBA.

D. Inference System

The inference system is the core of the proposed framework, which is designed for all of the calculation work of PCBA defect detection. To improve the flexibility and miniaturization of the AOI equipment, an inference system is deployed on an RK3399 Pro which is a lightweight artificial intelligence chip. The hardware architecture of the inference system is shown in Fig. 4. In terms of software, lightweight virtualization technology and container orchestrator are used to implement application service deployment and flexible configuration orchestration.

As mentioned before, in the inference system, original

images are processed in three steps which are image pre-processing, semantic segmentation, and defect detection, respectively. The entire inference process is shown in Fig. 4. In the pre-processing stage, a batch of images obtained in IAS are stitched together into a complete PCBA image denoted as I_0 , and redundant and overlapped areas are removed. Then the image I_0 is transmitted as input to the neural network. Encoded and decoded by the network, a segmentation feature map is outputted which is denoted as I_{seg} . The feature map consists of semantic information which divides the PCBA into three categories including background, soldering areas, and chips. Finally, detection is completed based on I_{seg} along with a ground truth template.

E. Evaluation Methods

Four main indicators are applied to evaluate the performance of the defect detection process, which are the mean intersection over union (mIoU), false-call rate (FCR), missing detection rate (MDR), and inference time (InfT), respectively. IoU is a primary evaluation indicator and reflects the proportion of correctly classified areas. The IoU value of defect class C_i can be calculated as follows:

$$IoU(C_i) = \frac{TP}{TP + FP + FN}. \quad (1)$$

Taking all the defect classes into consideration, mIoU is introduced to evaluate the comprehensive performance of the model.

$$mIoU = \frac{1}{k} \sum_{i=0}^{k-1} \frac{p_{ii}}{\sum_{j=0}^k p_{ij} + \sum_{j=0}^k p_{ji} - p_{ii}} \quad (2)$$

where p_{ij} represents the ratio of defects in type i predicted as type j . k is the number of defect classes, where the value of k is 5.

On the basis of weak links in actual PCBA production, FCR and MDR are proposed as two significant indicators to evaluate the performance. FCR focuses on those areas which are mistakenly detected as defects. It is calculated as follows:

$$FCR = \frac{FN}{TP + FN + TN + FP}. \quad (3)$$

MDR represents the proportion of those defects which cannot be detected by the proposed method. It is calculated by

$$MDR = \frac{FP}{TP + FN + TN + FP} \quad (4)$$

where FP , FN , TP , and TN denote the false positive, false negative, true positive, and true negative, respectively. Negative means there is the existence of defects and positive refers to normal areas.

Both FCR and MDR reflect the model accuracy in detecting minor defects. Model parameters can be optimized based on these two indicators to improve PCBA production efficiency.

Inference time is another considerable indicator which evaluates the calculation efficiency of the model. InfT is supposed to be as small as possible to improve the efficiency of the production line.

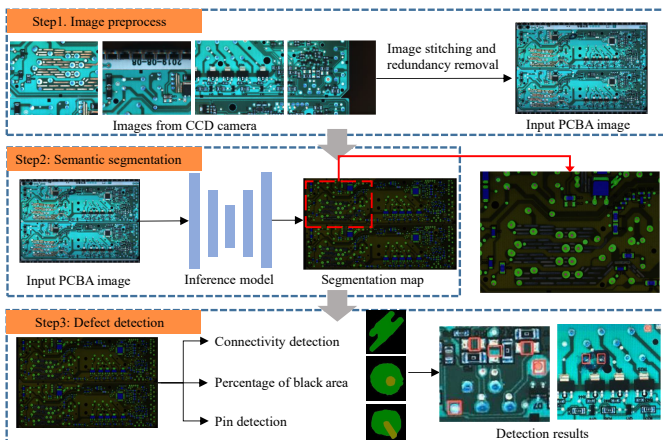


Fig. 4. Sketch of the proposed inference system.

IV. ALGORITHM DESIGN

A. Architecture Overview

Unlike conventional defect detection methods mentioned in Section II, our proposed method combines the advantages of both TI and TF methods, which is represented as a two-stage method. That is, a segmentation map is first obtained by a convolution neural network. This map contains semantic information of PCBA, in which all the soldering areas, chips, and pins are separated from the background and highlighted with different colors. Then in the second step, these segmented areas are compared with a standard contour map to localize the defected areas.

As shown in Fig. 5, the input image is in RGB format. As there are no complicated operators in U-Net, which is friendly for model deployment in EIS, we utilize U-Net as the backbone to convert input images into segmentation feature maps. To accelerate the training process and reduce the model size, several improvements are proposed including model quantization and shrinking base channels. Moreover, local residual blocks are used in the decoder stage to improve model performance. In another branch, a rule-based defect recognition algorithm is proposed to obtain the localization information of all the defects. With the assistance of a segmentation map, the defect region can be rapidly localized rather than determined with pixel-wise inspection.

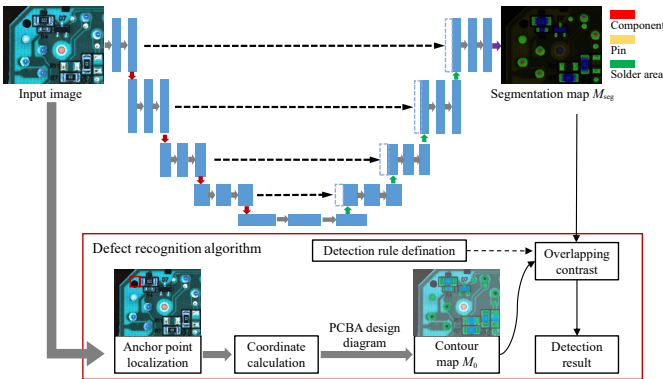


Fig. 5. Architecture of proposed defect detection method. This method combines the semantic segmentation network with rule-based detection algorithms to localize micro defects on PCBA.

B. Rule-Based Defect Recognition

To find the positions of all defects, we propose a simple but efficient defect recognition method which refers to the design diagram of PCBA. In this method, A contour map M_0 is firstly obtained based on the design diagram of PCBA and utilized as a benchmark. This contour map contains four regions including the soldering region, pins region, device region, and background region. One thing to be noticed is that region contour coordinates are calculated based on the anchor point on PCBA (shown in Fig. 5) and position information from the design diagram. During the inspection process online, the segmentation map M_{seg} extracted by the U-Net model overlaps with M_0 . A rule based on the overlapping area threshold and the principle of proximity is defined to judge whether a defect exists. Under this rule, we can recognize all the listed defects of the

PCBA to be inspected.

As shown in Fig. 6(a), we obtain the design diagram which contains the soldering and device regions (M_0) for the given PCBA image based on the design of PCBA. Similar to the segmentation map M_{seg} shown in Fig. 6(b), there are two device regions missing, which are marked with red boxes. Based on the rule, we can realize the components missing defect detection.

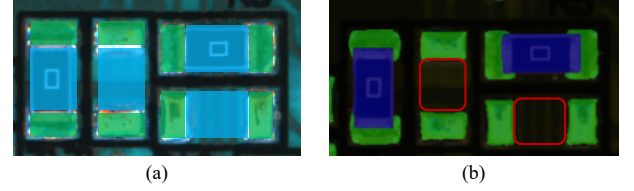


Fig. 6. The rule designed for detecting components missing defects. (a) The design diagram for a given PCBA; (b) The components missing defects are detected by the rule-based algorithms.

C. Group Convolution

Group convolution originates from AlexNet in 2012. In a deep convolution network, a convolution kernel $H \times W \times D$ can be regarded as a three-dimensional filter including the channel dimension D and space dimension (H, W) . And there is an assumption that the correlations between kernel channels can be decoupled. Thus, we can divide the original kernel with D channels into N individual kernel groups and map them with N different input feature maps respectively. Then, N output feature maps are obtained and each has D/N channels. These feature maps are loosely coupled and can be concatenated in the dimension of the channel, forming an ultimate feature map with D channels. In this way, the number of total model parameters is reduced by $1/N$, which improves the calculation efficiency. Moreover, with the assistance of group convolution, the training process of the model can be accelerated by distributing them on multiple GPUs for parallel computing.

D. Model Quantization

Considering the total amount of parameters, the size of the segmentation model reaches as big as 28 Mb, which constrains the computational efficiency of the inference system. INT8 quantization, as one of the methods to accelerate computing for deep convolution models, is utilized here to improve the model efficiency. The data type of weights is converted from float 32 to int 8, which reduces the computational requirements by one quarter.

E. Residual Block

To improve the performance of the segmentation map, we propose a residual block structure in each convolution block of the decoder stage. As shown in Fig. 7, after each upsampling process, the feature map I_{dec} is concatenated with the feature map I_{enc} which is copied from the corresponding layer in the encoder stage, forming a merged feature map denoted as $I_{dec+enc}$. Then, $I_{dec+enc}$ is calculated by two convolution layers and converted to feature map $I_{dec+enc}^{(2)}$. In the conventional U-Net model, $I_{dec+enc}^{(2)}$ is usually directly processed by up-sampling. However, in our approach, $I_{dec+enc}^{(2)}$ is added

with I_{dec} before up-sampling and this operation is applied in all the convolution blocks of the decoder stage. In this way, the degeneration of the network can be alleviated and the convergence rate is improved during the training process of the model.

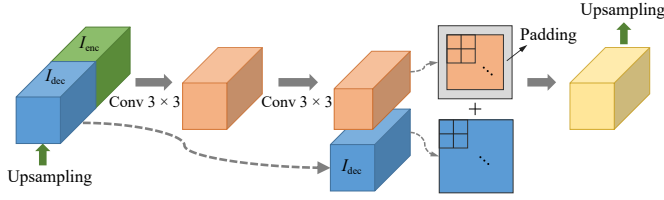


Fig. 7. Residual block used in the decoder stage. This figure only shows the utilization of residual block in a single convolution block of the decoder stage. It is applied in all the convolution blocks in the decoder stage. Here, I_{enc} and I_{dec} are the input from the encoder and decoder in U-Net correspondingly.

V. EXPERIMENT AND RESULTS

A. Datasets

The sample data utilized to train the proposed model is obtained by a real AOI system, which works in the same mode as that mentioned in 3.3. That is, 20 cropped images are obtained from one PCBA in a detection cycle. A total of 980 images are collected with a resolution of 2448×2048 . All these PCBAs contain defects of varying types which are listed in Table I. As there are no public PCBA datasets for image segmentation, so we construct our own dataset for validation.

With data annotation, we divide all the elements of the PCBA image into 4 categories which are the soldering joints, pins, electronic components, and background, respectively, which we think it is sufficient to distinguish the components of SMT. Here, we utilize the labeling tool “labelme” to annotate the data in the Pascal VOC data format.

Three of these categories mentioned above are labeled manually except for the background. In the process of labeling data, we draw the outlines of each element and fill different categories with different colors. For defects detection, we can label more categories with the same rule to expand the training datasets.

B. Training and Evaluation

We use the labeled data to train the proposed model which is implemented in Python 3.6. The model is built based on Pytorch 1.7 and trained on a cloud server consisting of NVIDIA Tesla V100 GPUs. We randomly chose 90% of the sample data as a training set and the rest 10% as a test set. 160 000 iterations are set to train the model with a batch size of 8. During the training process, the initial value of the learning rate is set to be 9.997×10^{-3} and it follows a decay rate of 0.0005 in each epoch. Cross entropy loss function is used in the training which is represented by

$$loss = - \sum y_{true} \log(y_{pred}) \quad (5)$$

where y_{true} denotes the true category number (0, 1, 2, and 3 which represent soldering joints, pins, electronic components,

and background, respectively) of each pixel, and y_{pred} is the segmentation result. All the pixels in an image are involved to calculate the loss value. Based on the rule-based defect recognition mentioned in Section IV, we can detect the defects with y_{pred} and validate with y_{true} .

Among all the hyper-parameters used in model training, base channels, batch size, and learning rate are the most influential parameters. To improve the training performance of the proposed model, we compare the impacts of different hyper-parameters on the training process and try to find an optimal parameter combination. As shown in Fig. 8, the loss curves of different training processes show that the proposed model represents the best convergence speed when the base channel is 16, batch size is 64, and the learning rate policy is poly.

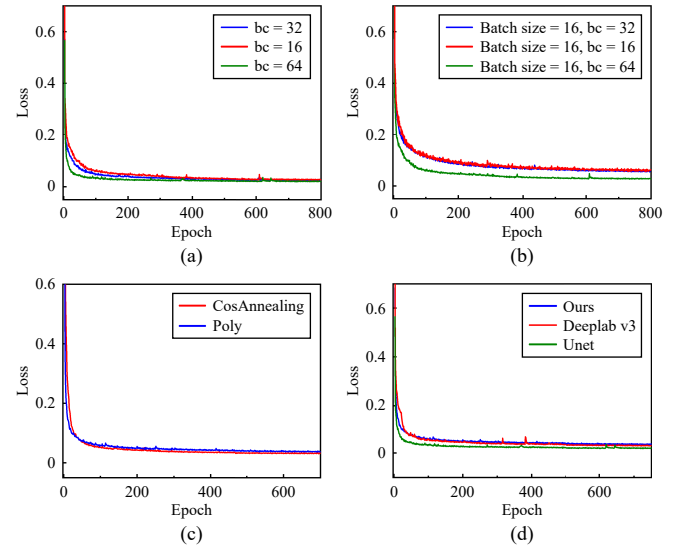


Fig. 8. Loss curves of different training processes. (a) Loss curves obtained under different base channels; (b) Comparison of loss curves under different batch sizes; (c) Loss curves under two different policies of learning rate; (d) Training processes of three different segmentation models including U-Net, deeplab-v3, and ours.

To show the performance of our model, training processes with other models are conducted for reference, in which traditional U-Net and deeplab-v3 are utilized as the backbone. As shown in Fig. 8, the loss of our method converges more rapidly than the other models. Furthermore, among all three models, the segmentation map output by our proposed model has the highest *mIoU* value, showing its advantage in extracting the semantic information of input images.

As mentioned in Section IV, some approaches are proposed to accelerate the training process and improve the model performance, including group convolution and residual block approaches. Benefiting from these improvements, our model can accurately recognize the hundred-micron-scale features and identify the millimeter-scale devices in PCBA, which can be applied to most types of PCBA with different complexity. Fig. 8(d) shows the effectiveness of these approaches through a comparison between standard U-Net and our model. Meanwhile, the *mIoU* value can also prove that our model gets better segmentation performance with these improvements.

C. Experiment and Results Experiment Setup

To validate the performance of the proposed method in industrial applications, we deploy the detection model in RK3399 Pro and build up a test system based on the architecture introduced in Section III-B. As shown in Fig. 9, to make the test system reliable and convincing, we choose the same image acquisition equipment and production line as in actual manufacturing and simulate a real PCBA defect detection process. There is a slight difference in this test system in that we replace some irrelevant process links with two robotic arms denoted as R_1 and R_2 . R_1 and R_2 simulates the processes before and after defect detection, respectively. In this way the production line is simplified and working efficiency is improved without affecting the confidence of the experiment. 98 PCBA samples from the actual production line are inspected in the test system. These PCBAs are slightly different in layout from the training datasets mentioned before. Each PCBA has a random number of defects ranging from 0 to 7.

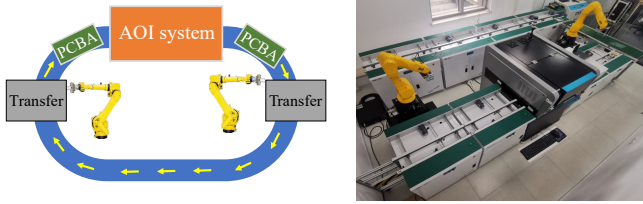


Fig. 9. Test system aimed at validating the performance of the proposed method. In experiment 1, the inference system is not utilized and the detection is accomplished using the traditional XOR method embedded in AOI equipment. In experiment 2 our proposed method is introduced and deployed in a lightweight computing device.

D. Results and Discussion

In actual PCBA production, throughput is one of the major factors which need to be focused on. Throughput is highly influenced by the detection process, which is a major limitation variable in the whole SMT line. As mentioned before, detection performance is measured by indicators such as FCR, MDR, and InfT. As shown in Table II and Fig. 10, we compare the values of different indicators in a 98-PCBA-detection process before and after using the proposed method. Initially, we use a traditional XOR algorithm embedded inside of the AOI to detect the PCBAs. In this method, the MDR of the detection process is 0.5% and FCR reaches to as high as 5%, which is counted from the level of soldering points. In a single detection process, as many as 10 soldering areas are reported as defects which are later judged to have passed once they are reviewed by an experienced operator.

After using the proposed method, we have witnessed a sharp decrease in FCR and MDR, which drop to 0.04% and 0 respectively. The maximum number of false areas in a single PCBA has reduced from 10 to 2. According to tests with experienced operators, the reduction of FCR can eliminate at least 80% of manual work spent on reviewing alerted PCBAs. Meanwhile, compared with the traditional detection method, detection speed has been reduced from 20 s per image to 6 s per image. Considering the complexity and high resolution of the PCBA, the detection speed is high enough for actual pro-

TABLE II
PARAMETERS OF DETECTION PERFORMANCE

	Traditional XOR method	Our method
FCR	5%	0.04%
MDR	0.5%	0
InfT (img/s)	20	6
Throughput (h^{-1})	180	600
Power consumption (w)	About 200	14

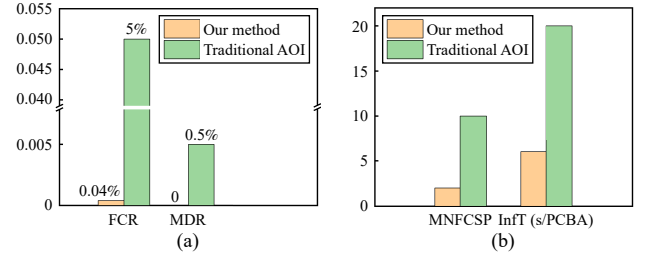


Fig. 10. Detection performance of traditional AOI (blue) and AOI with our method (orange). (a) MDR (left axis) and FCR (right axis); (b) MNFCSPP (left axis) and InfT (right axis).

duction. As a result, the throughput of the PCBA production line has been improved from 180 h^{-1} to 600 h^{-1} , which shows a great improvement in production efficiency. Moreover, the power consumption is also a significant indicator to evaluate the performance of defect detection, especially considering long-term and large scale production. In order to show the power consumption spent on defect detection, we compare our method with traditional XOR and object detection (OD) methods under the same working conditions. Traditional methods, whether they are XOR with a template or OD with deep neural networks, have to be run on industrial personal computers (IPCs) because of the size of their algorithms. According to IPC operating data during detection, the power can reach to about 200 w. While our method consumes only 14 w with the help of integration into EIS.

To look further, the proposed method, especially the semantic segmentation model, has to be adjusted based on production commands before it adapts to the production line. In PCBA production lines, throughput and energy consumption are two significant objectives. As mentioned before, throughput is influenced by detection accuracy and inference time. Therefore, in order to optimize detection accuracy, we adjust the base channels of the U-Net model to promote the IoU value and collect enough real PCBA datasets to ensure the generalization capability of the model. Meanwhile, in order to reduce inference time, we utilize model quantization along with group convolution to deflate the model, which can also make the model feasible for lightweight computing devices. In summary, we design and implement a general adjustment policy for the PCBA detection model based on actual production objectives.

VI. CONCLUSION

In this paper, a lightweight PCBA defect detection method based on a deep convolution neural network is proposed. Firstly, we use a semantic segmentation model combined with

computer vision methods to build up a defect detection framework. Extensive real PCBA images from actual production lines are collected and annotated as datasets, which makes the proposed model generalize to new PCBA structures.

Moreover, optimization methods such as model compression and local residual blocks have been applied in the segmentation model according to production demands. This enables the proposed method integrated into lightweight computing devices and promotes inference time from 20 s/img to 6 s/img. Experiment results from actual PCBA production lines show that the proposed method can effectively improve the detection accuracy at least by one order of magnitude and push the throughput from 180 h⁻¹ to 600 h⁻¹. Thus, the computing consumption has been reduced to 14 w.

Finally, by training and testing experiments under different conditions, we design and implement a general paradigm for optimizing the PCBA detection model based on actual production objectives. With the increase of PCBA complexity, this paradigm can also provide us with a guide for adapting the detection model to new production lines.

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Yang Li received the B.S. degree in mathematics and applied mathematics and M.S. degree in mathematical statistics from Nankai University in 2014 and 2017, respectively. He is currently a Researcher with the Department of Internet of Things Technology and Application, China Mobile Research Institute. His research areas include machine learning, computer vision, and the Internet-of-Things.



Xiao Wang received the B.S. and M.S. degrees in mechanical engineering from Xi'an Jiaotong University in 2013 and 2016, respectively, the Ph.D. degree in mechanical engineering from Tsinghua University in 2021. He is currently a Researcher with the Department of Internet of Things Technology and Application, China Mobile Research Institute. His research interests include machine learning (especially deep neural networks), industrial measurement technology, computer vision and Internet-of-Things.



Zhifan He received the B.S. degree in physics from Fudan University in 2014, the M.S. degree in integrated circuit engineering from Tsinghua University in 2023. He is currently an Engineer at the SigmaStar Technology Limited. His researches focus on the hardware-software co-design and development for in-memory computing in heterogeneous systems.



Ze Wang received the B.S. degree in information engineering from the College of Optical Science and Engineering, Zhejiang University in 2011, the M.S. degree in electrical and computer engineering from Purdue University in 2022. Currently, he is currently a Ph.D. candidate in integrated circuit engineering at the School of Integrated Circuits, Tsinghua University. His research interests include in-memory computing technologies and artificial intelligence chips.



Yawen Niu received the M.S. degree in computer science and technology from Tsinghua University in 2006. He is currently the Vice Director of the Department of IoT Technology and Application, China Mobile Research Institute, leading research and development of new IoT technology and advanced computing technology. His research interests include new IoT technology and advanced computing technology.



memory computing technologies and artificial intelligence chips.

Ke Cheng received the B.S. degree in electronic information science and technology from the School of Information Science and Engineering, Southeast University in 2001 and the M.S. degree in signal and information processing from the National Key Laboratory of Mobile Communications, Southeast University in 2004. He is currently a Ph.D. candidate in integrated circuit engineering at the School of Integrated Circuits, Tsinghua University. His current research focuses on the cutting-edge fields of in-



Shapeng Xiao received the M.S. degree in engineering of telecommunication from the Beijing University of Posts and Telecommunications in 2001. He is currently the Director of the Department of IoT Technology and Application, China Mobile Research Institute, leading research and development of new IoT technology, industry internet, and vertical digitalization. His research interests include new IoT technology, industry internet, and vertical digitalization.



Sanchuan Ding received the B.S. degree in electronic science and technology from Tianjin University in 2005, and the M.S. degree in microelectronics and solid state electronics from Tianjin University in 2007. Currently, he serves as the Director of the Department of Hardware and Systems at the Beijing Advanced Innovation Center for Integrated Circuits, Tsinghua University. His research interests include novel memory devices and brain-inspired computing technologies.



Zhenqi Hao received the B.S. degree in mathematics and physics and the Ph.D. degree in condensed matter physics from Tsinghua University in 2013 and 2018, respectively. He is currently working with Beijing Advanced Innovation Center for Integrated Circuits at Tsinghua University. His research interests include novel memory devices and brain-inspired computing technologies.



Yijing Fan received the M.S. degree in information and communication engineering from Beijing University of Posts and Telecommunications in 2022. She is currently a Researcher of the Department of Internet of Things Technology and Application, China Mobile Research Institute. Her research interests include machine learning (especially deep neural networks), industrial measurement technology, computer vision and Internet-of-Things.



Bin Gao (Senior Member, IEEE) received the B.S. degree in physics and the Ph.D. degree in microelectronics from Peking University in 2008 and 2013, respectively. He is currently an Associate Professor with the School of Integrated Circuits, Tsinghua University. His research interests include fabrication, characterization, and modeling of emerging semiconductor devices, especially RRAM.



Xiaotao Li received the B.S. degree in automation from Dalian Maritime University in 2009, the M.S. degree in control theory and control Engineering from Dalian University of Technology in 2012, and the Ph.D. degree in pattern recognition and intelligent system from Beihang University in 2016. He is currently a Senior Engineer with China Mobile Research Institute. His research interests include machine intelligence, knowledge graph, and the Internet of Things.



Huaqiang Wu (Senior Member, IEEE) received the B.S. degree in material science and engineering and enterprise management from Tsinghua University in 2000, and the Ph.D. degree in electrical engineering from Cornell University, USA in 2005. From 2006 to 2008, he was a Senior Engineer with Spansion Limited Liability Company, USA. In 2009, he joined the Institute of Microelectronics, Tsinghua University, as an Associate Professor. He was promoted to Full Professor in 2018. He is currently the Director of the School of Integrated Circuits, Tsinghua University. His research interests include advanced memory and brain-inspired computing technologies.